

AIMO Proof Pilot — Final Grade

Problem (#_ID): 4_COLOUR

Submission: model_2

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The solution states the correct answer and has the right high-level idea (designating one cell in four as “black”), but both of its strategies are flawed. Linda’s strategy is significantly wrong: it only paints a black cell once it is “safe”, which is far too conservative — a black cell can be overpainted by Sue yet still have grey neighbours, so Sue can avoid ever making it safe (the claim that a black cell’s last Sue move makes it safe is false). The lower-bound argument therefore fails. Sue’s strategy also has genuine gaps (the last row/column handling is not rigorous, the fixed block order can prescribe an illegal move, and the count implicitly assumes Linda only colours black cells).

However, the essential idea of Sue’s argument — greedily covering the board with 2×2 blocks and counting the moves — does establish that the number of lime cells is at most B , which is the partial-progress milestone **PP2**. Although this direction is not cleanly explained, it is enough to earn the partial mark. The solution therefore scores **1**.